Seminar 2

(Prepare 3 questions)

Operational Semantics

1. What is **operational semantics**?

2. How does operational semantics differ from **axiomatic** semantics?

3. Why do programming language designers use operational semantics to describe a language?

4. What is the meaning of a **<Expr, State> = Value** in operational semantics?

5. What is the meaning of a **<Command, State1> → State2** in operational semantics?

6. What does the **skip** command represent? Explain in plain words the meaning of the rule

   <skip, State> → State

7. Explain in plain words the meaning of the operational semantic rule for

   <Command1 ; Command2, State>

8. Explain in plain words the meaning of the operational semantic rule for the **if-then-else** command.

9. Explain in plain words the meaning of the operational semantic rule for the **while** command.

10. Specify the operational semantics for the **SWITCH** statement, e.g.

    SWITCH (n) case n1: command1; case n2: command2; DEFAULT: command3)

11. Describe the operational semantics of the following program fragment. Show the applications of all the rules. Assume **x=0** initially.

```
x := 2;
while x>0 do
  x := x-1.
```

12. Prove that the program **skip ; command** is semantically equivalent to **command**